

Project 1: Readmission Risk + Discharge Note Understanding (Tabular + Text)

Goal

Predict 30-day readmission from a small, synthetic EHR table (demographics, diagnoses, labs) **and** use discharge notes to add signal. Compare:

- Classical ML on tabular only (LogReg/XGBoost) vs.
- Fusion models that incorporate **text embeddings** (BioClinicalBERT or general SBERT) into the tabular model.

Data (create locally; ~10–20k patients)

- patients.csv (id, age, sex, comorbidity_index, los_days, num_prev_adm, primary_dx, avg_hr, avg_creatinine, readmitted_30d)
- notes/ID.txt synthetic discharge notes with patterns matching the tabular risk (e.g., high comorbidity → mentions “multiple chronic conditions”, etc.).

How to create/download

1. **Synthetic tabular** (use provided script)
2. **Synthetic notes**: template-based generation with controlled vocab tied to risk factors.

Starter code (dataset generation & baseline)

Separately shared

What to compare

- **Tabular-only** vs **fusion** (tabular + SBERT/BioClinicalBERT embeddings)
- Try **spaCy** for NER features (e.g., problems, procedures) and **gensim** for topic proportions as extra features.

Evaluation rubric (100 pts)

- Translate business problem to an optimal technical solution, including approach, options explored, logical sequencing of the workflow and tasks involved: **20**
- Text processing & embedding integration (spaCy/gensim/transformer): **25**
- Baselines & comparisons with clear metrics (ROC-AUC/PR-AUC, calibration): **25**
- Presentation clarity: **10**
- Teamwork: **10**
- Timely completion and submission: **10**